

Q1 Harshad number

10 marks

A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: $18 \rightarrow \text{digit sum} = 9, 18 \% 9 == 0$. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500.

Your Answer

```
n=int(input())
s=sum(int(i) for i in str(n))
if n%s==0:
    print("Not Harshad Number")
for i in range(1,501):
    t=sum(int(j) for j in str(i))
```

The following Python program is intended to find the GCD (Greatest Common Divisor) of two numbers using the Euclidean algorithm. It contains FOUR errors. Identify each error and write the corrected line.

```
a = int(input("Enter first: ")) b = int(input("Enter second: "))
while b = 0: temp = a a = b b = temp % a
print("GCD is", a)
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 a=int(input("Enter first: "))
2 b=int(input("Enter second: "))
3 while b!=0:
4     temp=a
5     a=b
6     b=temp%b
7     print("GCD is", a)
```

Test Input

48
18

Output

```
Enter first: Enter second: GCD is
18
GCD is 12
GCD is 6
```

```
seen = _____ pairs = []  
for num in _____:  
    complement = target - _____  
    if complement in _____:  
        pairs.append((complement, num))  
    _____add(num)  
return _____  
data = [int(x) for x in input("Enter numbers: ").split()]  
t = int(input("Target: "))  
result = _____(data, t)  
print("Pairs:", result)
```

Your Answer

1. set()
2. lst]
3. num
4. seen
5. seen
6. pairs

Q4 Bitwise Operators

10

Trace the following Python program step by step. Write the exact output. Also state the data type of each variable after assignment.

```
a = 0b1101
```

```
b = 0o25
```

```
c = 0x1A
```

```
result = (a & b) | c
```

```
print(f"a={a}, b={b}, c={c}, result={result}")
```

```
print(bin(result), oct(result), hex(result))
```

```
print(result >> 2, result << 1, ~result)
```

Your Answer

```
a=13(int)
```

```
b=21(int)
```

```
c=26(int)
```

```
result=31(int)
```

Your Fixed Code

```
1 def rotated_binary_search(arr, target):
2     low = 0
3     high = len(arr) - 1
4     while low <= high:
5         mid = (low + high) // 2
6
7         if arr[mid] == target:
8             return mid
9
10        if arr[low] <= arr[mid]:
11            if arr[low] <= target < arr[mid]:
12                high = mid - 1
13            else:
14                low = mid + 1
15        else:
16            if arr[mid] < target <= arr[high]:
17                low = mid + 1
18            else:
19                high = mid - 1
20
21    return -1
```

Test Input

1

Output

Target: Found at: 5